

AD - 303 (GS)

B.Tech 111 semester

Examination, December 2022

Grading System (GS)

Data Structure

Time : 3 Hours Maximum Marks : 70

Note : i) Attempt any five questions .

ii) All questions carry equal marks .

1a) Explain Red Black Trees , also discuss the properties of Red Black Tree .

**~~b) Sort using quick sort algorithm –~~
36 , 25 , 32 , 5 , 8 , 65 , 38 , 47 , 95 .**

2a) What is Hashing ? Explain different Hash function in detail .

~~b) Explain AVL Tree . Insert the following elements in AVL Search Tree .~~

48 25 10 5 7 3 28 20 8 16

3 . a) Explain analysis of Heap Operation with example .

~~b) Sort the following list using quick sort .~~

44 , 33 , 11 , 55 , 77 , 90 , 40 , 60 , 99 , 22 , 88 , 66 .

4 . a) Write the algorithm of insertion sort and find its Time complexity .

b) Difference Between Internal sorting and External sorting .

5 . a) Give the solution for following Recursion – $T(n) = 2T[n/2] + n \log n$.

~~b) Difference between Breadth First Search and Depth First Search~~

6 . a) Write an algorithm for insertion and deletion in priority Queues .

b) Explain Dijkstra's algorithm for finding shortest path with an example .

7 . a) write short note on Internal trees . Explain with example .

b) How can you convert an infix expression to postfix operation using stack . Give one Example .

8 . a) Write an algorithm Which counts the number of connected component in a graph .

~~b) Construct a Tree for the given Inorder And Postorder traversals .~~

Inorder : DGBAHEICF

Postorder: GDBHIEFCA